

1. Administrative & Study Identification

Full Study Title: A Phase III, Double-Blinded, Multicenter, Randomized Study Evaluating the Efficacy and Safety of Neoadjuvant Treatment With Atezolizumab or Placebo in Combination With Platinum-Based Chemotherapy in Patients With Resectable Stage II, IIIA, or Select IIIB Non-Small Cell Lung Cancer **Short Title/Acronym:** IMpower030 **Protocol Number:** G040241 **NCT Number:** NCT03456063 **Sponsor Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Atezolizumab (Tecentriq) + Platinum-Based Chemotherapy (Nab-paclitaxel, Pemetrexed, Carboplatin, Cisplatin (Platinol), Gemcitabine) **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of neoadjuvant treatment with atezolizumab in combination with platinum-based chemotherapy versus placebo plus chemotherapy in patients with resectable Stage II, IIIA, or select IIIB Non-Small Cell Lung Cancer (NSCLC). **Secondary Objectives:** To evaluate Pathological Complete Response (pCR), Major Pathological Response (MPR), Objective Response (OR), Overall Survival (OS), Disease-Free Survival (DFS), and overall safety profile.

2.2 Background & Rationale To address the need for improved long-term outcomes and surgical resectability in patients with early-stage (Stage II-IIIB) NSCLC. By integrating immune checkpoint inhibition (atezolizumab) with standard neoadjuvant chemotherapy, the study aims to eradicate micrometastatic disease early and improve Event-Free Survival (EFS) and pathologic responses prior to curative intent surgery.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blinded **Randomization:** Randomized (1:1) **Trial Status:** Active, not recruiting

3.2 Planned Enrollment & Duration Target \$N\$: 453 participants **Number of Sites:** Global (United States + 23 other countries) **Estimated Study Start:** 24-Apr-2018 **Estimated Primary Completion:** 01-Oct-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years with an Eastern Cooperative Oncology Group (ECOG) performance status 0 or 1. 2. Histologically or cytologically confirmed, resectable Stage II, IIIA, or Select IIIB (T3N2 only) NSCLC of squamous or non-squamous histology. Staging should be based on the 8th edition of the AJCC/UICC staging system. 3. Evaluation by an attending thoracic surgeon to confirm eligibility for an R0 resection with curative intent. 4. Adequate pulmonary and cardiac function to undergo surgical resection. 5. Measurable disease as defined by RECIST v1.1. 6. Adequate hematologic and end organ function. 7. Negative HIV test at screening. 8. Negative for active HBV and HCV at screening. 9. Adequate tissue for PD-L1 IHC assessment.

4.2 Exclusion Criteria 1. NSCLC with histology of large cell neuroendocrine carcinoma or sarcomatoid carcinoma, or mixed NSCLC and small cell lung cancer histology. 2. Non-squamous NSCLC histology with activating ALK and EGFR mutations. 3. Any prior therapy for lung cancer. 4. Malignancies other than NSCLC within 5 years prior to randomization, with the exception of those with a negligible risk of metastasis or death treated expected curative outcome. 5. Pregnant or lactating women. 6. History of autoimmune disease. 7. History of idiopathic pulmonary fibrosis, organizing pneumonia, drug-induced pneumonitis, or evidence of active pneumonitis on screening chest Computed Tomography (CT) scan. 8. Prior treatment with cluster of differentiation 137 (CD137) agonist or immune checkpoint blockade therapies, anti-programmed-death-1 (anti-PD-1), and anti-PD-L1 therapeutic antibody. 9. Severe infection within 4 weeks prior to randomization. 10. Significant history of cardiovascular disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Atezolizumab 1200 mg IV on Day 1 of each 21-day cycle (every 3 weeks) for 4 cycles during the neoadjuvant phase, followed by adjuvant Atezolizumab 1200 mg IV every 3 weeks for 16 cycles. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** 4 neoadjuvant cycles (12 weeks) prior to surgery, plus 16 adjuvant cycles post-surgery (approx. 48 weeks).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications. **Prohibited:** Prior treatment with CD137 agonists, anti-CTLA-4, anti-PD-1, or anti-PD-L1 therapeutic antibodies; live attenuated vaccines within 4 weeks prior to initiation.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor biopsy for PD-L1, Baseline Scans
Neoadjuvant	Every 3 Weeks	Atezolizumab/Placebo + Chemotherapy infusion, Safety Labs (4 cycles)
Surgery	Post-neoadjuvant	Surgical resection (R0 intent), Pathological assessment (MPR/pCR)
Adjuvant	Every 3 Weeks	Adjuvant Atezolizumab (up to 16 cycles) or Best Supportive Care
End of Study	Follow-up	Long-term OS/EFS Follow-up, Survival Status, Safety monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Independent Review Facility (IRF)-Assessed Event Free Survival (EFS): IRF-assessed EFS is defined as the time from randomization to the first documented disease progression per RECIST v1.1 that precludes surgery, local or distant disease recurrence, or death from any cause, whichever occurs first. **Measurement Method:** Central and local pathology laboratory assessment and RECIST v1.1 criteria.

7.2 Secondary & Exploratory Endpoints * Pathological Complete Response (pCR) (absence of any viable primary tumor cells). * Major Pathological Response (MPR). * Objective Response (OR). * Overall Survival (OS) and Disease-Free Survival (DFS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via clinical assessments, laboratory monitoring, and patient reports. **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness; monitoring for life-threatening events or those requiring hospitalization. **Data Monitoring Committee (DMC):** External/Internal Data Monitoring Committee to regularly review safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for all patients receiving at least one dose of study drug. **Sample Size Justification:** \$N=453\$ adequately powered to detect a statistically significant improvement in IRF-assessed EFS and MPR. **Handling of Missing Data:** Standard Kaplan-Meier survival analysis methods and censoring rules applied for EFS and OS.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring visits by the sponsor. **Audit Trail:** Validated electronic data capture (EDC) system with full audit trail and data clarification forms.

11. Study Identifiers & Governance

Primary ID: G040241 **Posting ID:** 1128789149213034440 **Registry Number:** NCT03456063 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** A Study of Neoadjuvant Atezolizumab Plus Chemotherapy Versus Placebo Plus Chemotherapy in Patients With Resectable Stage II, IIIA, or Select IIIB Non-Small Cell Lung Cancer (IMpower030) **Official Regulatory Title:** A Phase III, Double-Blinded, Multicenter, Randomized Study Evaluating the Efficacy and Safety of Neoadjuvant Treatment With Atezolizumab or Placebo in Combination With Platinum-Based Chemotherapy in Patients With Resectable Stage II, IIIA, or Select IIIB Non-Small Cell Lung Cancer

12. Clinical Framework (Oncology Specific)

Primary Condition: Non-Small Cell Lung Cancer (NSCLC) **Diagnosis Threshold:** Resectable Stage II, IIIA, or Select IIIB (T3N2 only) **Medical Methodology:** Neoadjuvant chemo-immunotherapy followed by surgical resection and adjuvant immunotherapy **Optimization Target:** Major Pathologic Response (MPR) and Event-Free Survival (EFS)

13. Study Procedures & Methodology

Management Pathway: Standard oncology pathway for resectable NSCLC (Neoadjuvant -> Surgery -> Adjuvant) **Education Protocol (HCP):** Protocol training and RECIST v1.1 / Pathological evaluation guidelines **Education Protocol (Patient):** Informed consent, treatment scheduling, and post-operative care education **Quality Audit Mechanism:** Centralized pathology review for pCR and MPR; Independent Review Facility (IRF) for EFS

14. Observation & Follow-Up Schedule

Total Duration: Approx. 5-7 years (Including long-term survival follow-up) **Onsite Visit Cadence:** Every 3 weeks during the active treatment phase **Remote Monitoring Frequency:** Periodic survival follow-up **Checklist Review Interval:** Every 12 weeks during the follow-up phase for disease recurrence

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					